Database System Assignment

(CO 313)

A Project Report On

"Online Student Course Registration System for TU"

A Project Report Submitted

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INTRODUCTION

The title of the project is COURSE REGISTRATION SYSTEM. It aims at providing information to all the levels of admission within Tezpur University (TU). This system can be used as a registration in order to get admission in a course ordered by TU. For a given student a registration code is created and using this the administrator provide its result of the selected students under some given criteria. In the conventional File Processing System, the details of students are managed, a number of different application programs are written to extract records from and add records to the appropriate file. But this scheme has a number of major limitations and disadvantages such as data redundancy, data inconsistency, non-sharable data, non-standardized data, insecure data, incorrect data, etc. This results in an unsatisfactory registration system. The proposed system will have the capabilities allowing instructor the ability for creation, modification, migration, and archiving of courses by implementing the methods for reusability of the content.

PROBLEM DEFINATION

This document is the Software Requirement Specification (SRS) for the Course Registration System. The purpose of this document is to describe the functionality, requirements and general interface of the student. This document specifies analysis and design guidelines of Course Registration System. The specifications are intended to guide the group through the development process by which the matrix is measured and the success of the project is judged.

The goal of developing this model is to reduce the manual work of the Course Registration System which will therefore improve the overall working and the efficiency of Tezpur University.

PROJECT SPECIFICATION

A description of the project:

• Each student has a unique student ID and a profile. The profile includes first/last names, gender, date of birth, local address, department, enrolled year, username, login password, and may have a picture. You can also add other necessary info.

• Each instructor has a unique faculty ID and a profile. The profile must indicate the instructor's department(s). An instructor may work at more than one department.

• A department has a unique department ID, a name, and a list of faculties.

• Each course has a course code, an instructor, given department, term, credits, classroom, periods, prerequisite courses, textbooks, and other information you think is necessary.

• A classroom has a unique ID and a unique location. Classrooms can hold more than one course each term, but these courses cannot conflict in time. Classrooms have capacities. Registered students' number cannot exceed the classroom capacity.

• Must maintain all the courses a student has already taken/registered. This is used to check course

prerequisites when registration.

• Students may login (with username and password) to the system to register courses or retrieve all the courses they have already taken/registered.

• Instructors may login (with username and password) to the system to add courses or retrieve all the courses they have already given/added.

• A student cannot register a course if:
1) he/she doesn't meet the prerequisites,
2) the students registered in the course exceed the capacity of the classroom,
3) the course has a time conflict with other courses in the same term.

• An instructor cannot add a course if:
1) the classroom has already occupied by another course in the same period,
2) he/she has another course in the same period,
3) he/she is not affiliated to the department of this course.

• Students may retrieve all the courses given by a department, an instructor, or held in a specific period.

• There can be an arbitrary number of students/instructors/departments/classrooms/courses

Design

- **USE CASE DIAGRAM**

- **ER/EER model**

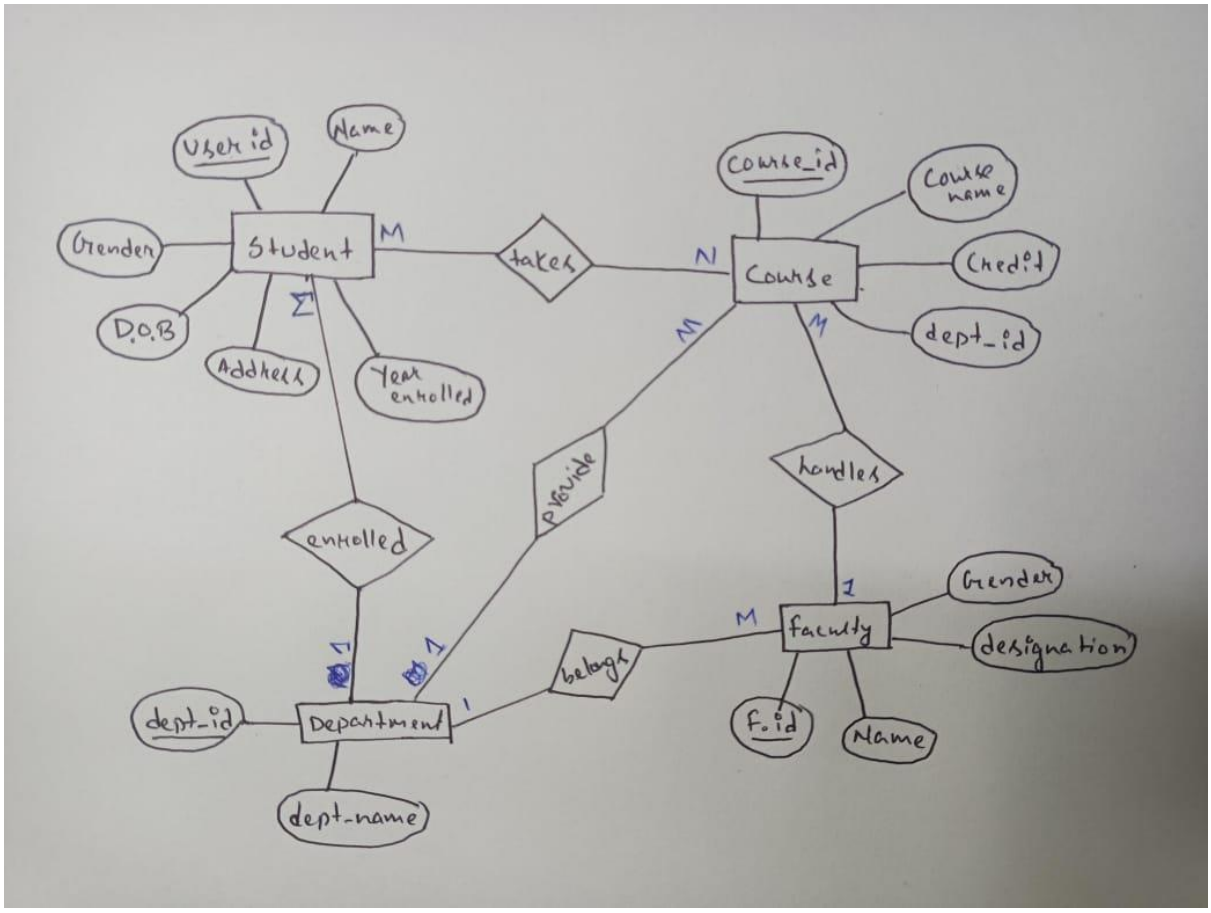


Fig: ER/EER Model for Online Course Registration System

The Entity-Relationship (ER) or Enhanced Entity-Relationship (EER) model is a framework used in online college course registration systems to manage course enrolment, student information, and administrative functionalities. The model consists of various entities, relationships, attributes, and interconnections, such as 'Student', 'Course', 'Instructor', 'Department', 'Semester', 'Registration', and 'Administrative User'. These entities are interconnected through relationships, such as 'Registration' indicating a student's enrolment in a specific semester, 'Instructor' indicating which instructor teaches which course, and 'Administrative User' allowing authorized personnel to manage student registrations and administrative tasks. Additional features like specialization hierarchies, inheritance, or subtypes can be incorporated in the EER model. The relationships between entities are defined by constraints and cardinalities, such as 'Course' having a many-to-many relationship with

'Student'. The ER/EER model serves as a blueprint for organizing and illustrating the structure, connections, and functionalities necessary for effective management and operation of the system.

- **Relational Model**

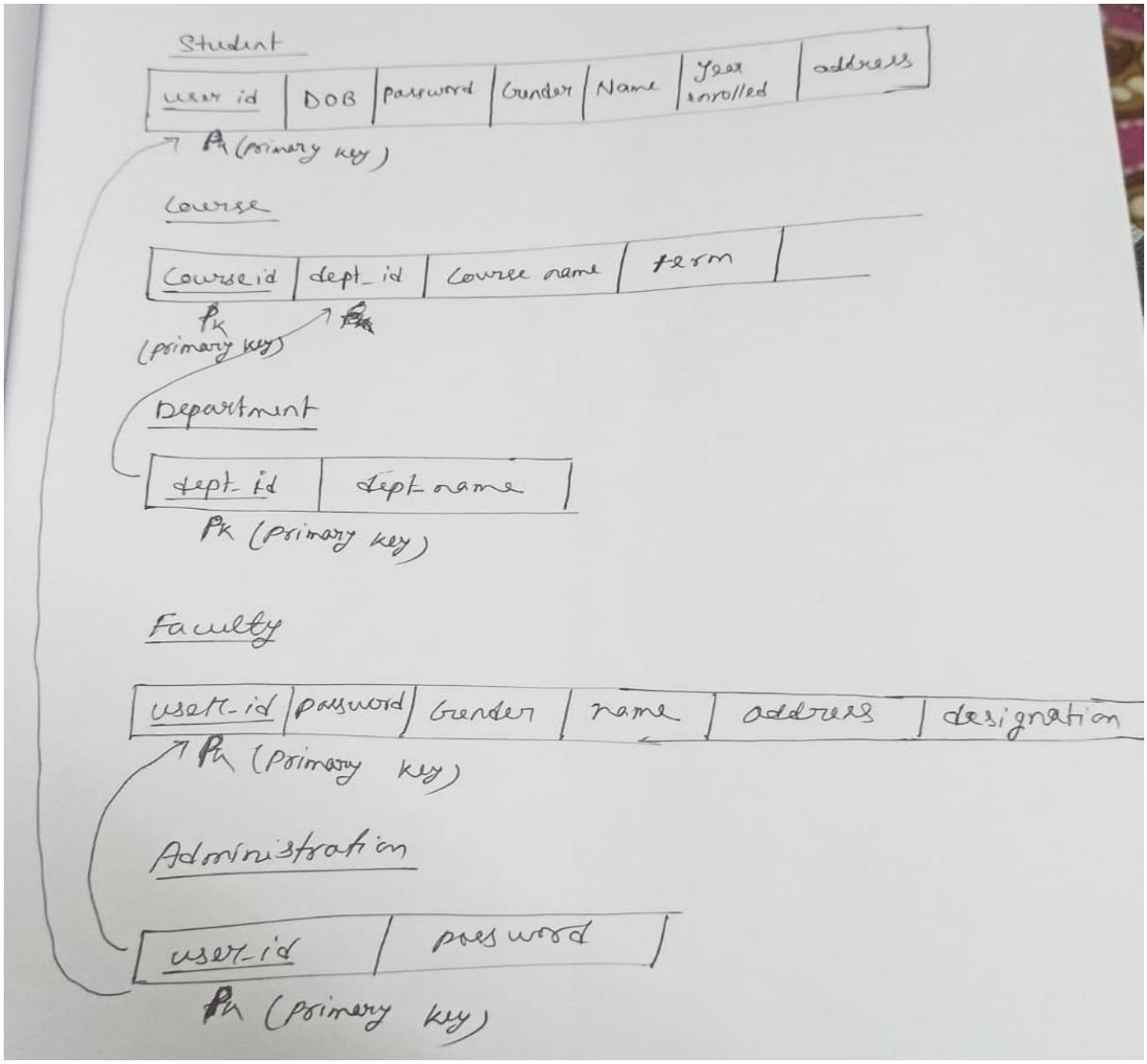


Fig: Relational Model for Online Course Registration for TU

IMPLEMENTATION DETAILS

i) Hardware Used

- Processor: AMD Ryzen 5
- RAM: 16 GB
- Storage: 512 GB

ii) Software Used

- Database: SQL
- Operating System: Windows 11
- Design and Coding Details: PHP
- Documentation Details: MS Word
- Browser: Brave
- Server: XAMPP

SQL commands used for creating the database

```
CREATE TABLE Student (
   User_ID INT PRIMARY KEY,
   Name VARCHAR(255),
   Date_of_Birth DATE,
   Address VARCHAR(255),
   Year_Enrolled INT,
   Gender VARCHAR(10)
);
CREATE TABLE Department (
   Dept_id INT PRIMARY KEY,
   Dept_name VARCHAR(255)
);
CREATE TABLE Course (
   Course_id INT PRIMARY KEY,
```

```
    Course_name VARCHAR(255),
    Dept_id INT,
    Credit INT,
    FOREIGN KEY (Dept_id) REFERENCES Department(Dept_id)
);
CREATE TABLE Faculty (
    F_id INT PRIMARY KEY,
    Name VARCHAR(255),
    Designation VARCHAR(255),
    Gender VARCHAR(10)
);
CREATE TABLE Enrollment (
    User_id INT PRIMARY KEY,
    Dept_id INT,
    FOREIGN KEY (User_id) REFERENCES Student(User_id),
    FOREIGN KEY (Dept_id) REFERENCES Department(Dept_id)
);
CREATE TABLE CourseDepartment (
    Course_id INT PRIMARY KEY,
    Dept_id INT,
    FOREIGN KEY (Course_id) REFERENCES Course(Course_id),
    FOREIGN KEY (Dept_id) REFERENCES Department(Dept_id)
);
CREATE TABLE FacultyCourse (
    F_id INT,
    Course_id INT,
    PRIMARY KEY (F_id, Course_id),
    FOREIGN KEY (F_id) REFERENCES Faculty(F_id),
    FOREIGN KEY (Course_id) REFERENCES Course(Course_id)
);
CREATE TABLE StudentCourse (
    User_id INT,
    Course_id INT,
    PRIMARY KEY (User_id, Course_id),
    FOREIGN KEY (User_id) REFERENCES Student(User_id),
    FOREIGN KEY (Course_id) REFERENCES Course(Course_id) )
```

Description of different modules of the system

1. User Authentication and Authorization:

- This module manages user accounts, login/logout functionalities, and verifies user identity.
- It ensures appropriate access levels based on roles (students, administrators) and permissions.

2. Course Catalog Management:

- Contains information about available courses, including course codes, titles, descriptions, prerequisites, credit hours, schedules, and instructors.
- Allows administrators to add, update, or remove courses.

3. Student Profile Management:

- Enables students to create and manage their profiles, including personal information, academic history, enrolled courses, and schedules.
- Allows students to update their contact information, view grades, and track their academic progress.

4. Course Registration:

- Provides students with the ability to browse the course catalog, search for specific courses, and add/drop courses within specified enrollment periods.
- Validates prerequisites and constraints to ensure eligibility for course registration.

5. Schedule Generation:

- Generates student schedules based on the selected courses, avoiding time conflicts.
- Provides options to view, edit, and finalize schedules before confirmation.

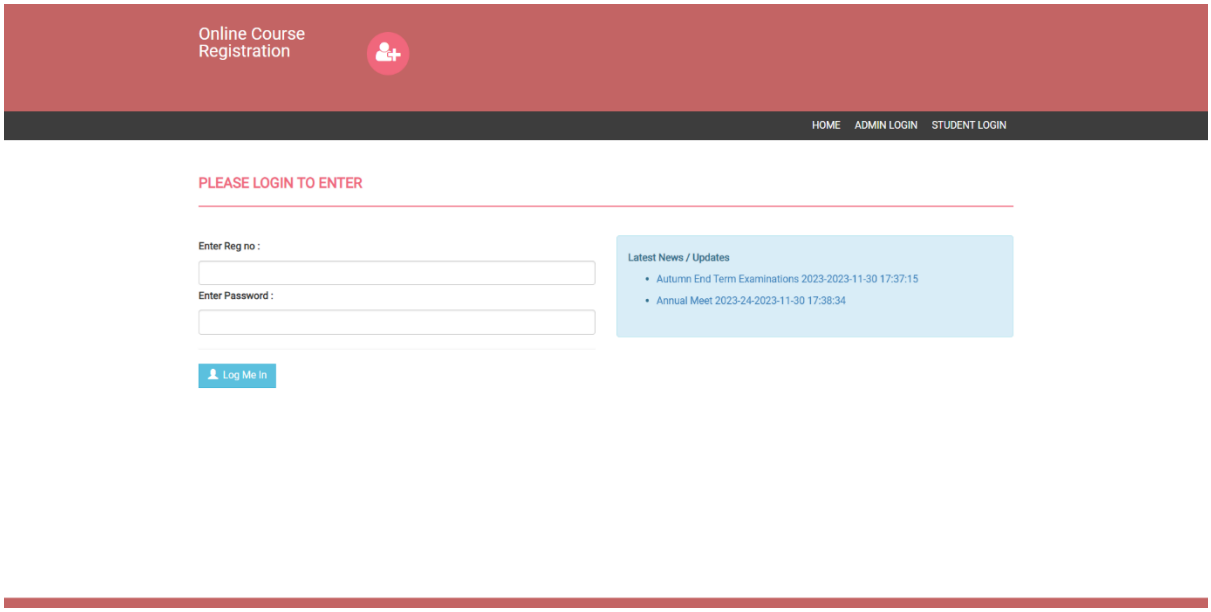
6. Administrative Dashboard:

- Gives administrators access to oversee the entire system, manage user accounts, handle exceptions, and resolve issues.
- Provides tools for monitoring enrollment statistics, generating reports, and managing system configurations.

7. Notifications and Alerts:

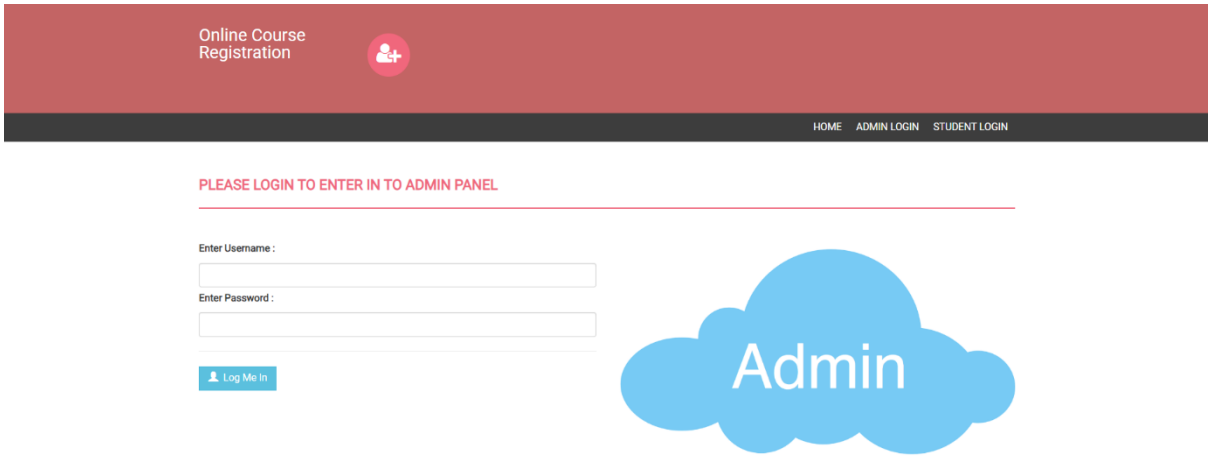
- Sends automated notifications to students regarding important dates, registration deadlines, course changes, and other relevant information.
- Notifies administrators about system updates, errors, or issues that need attention.

Snapshots of input/output forms and reports (till date)

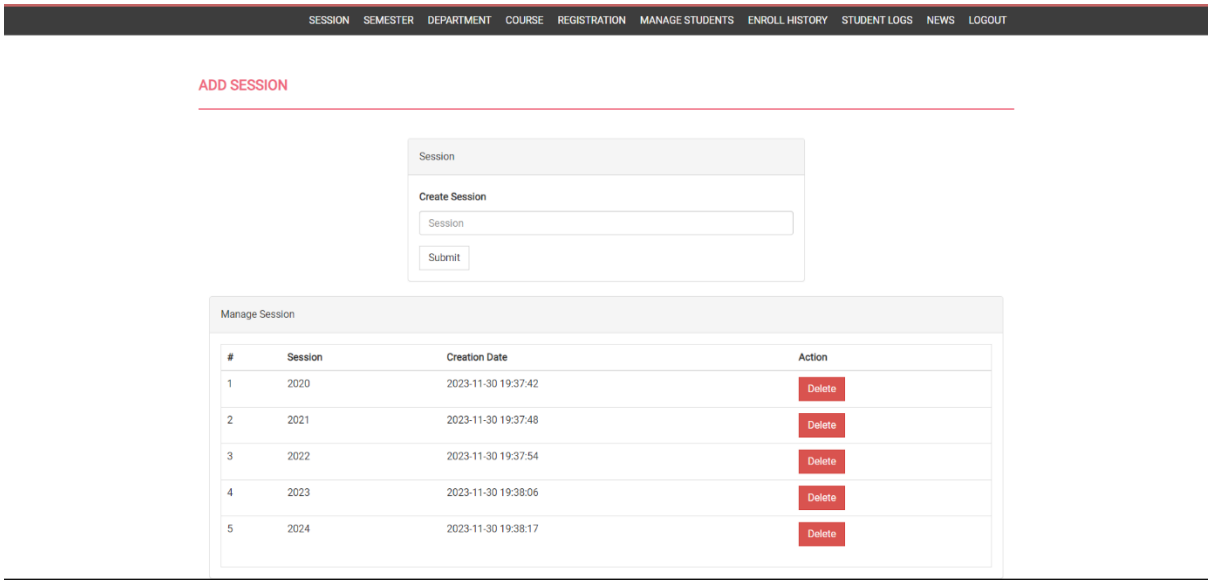


It is the home page/ Login Page of the table. Here you can choose in which form you login i.e., in form of admin or student

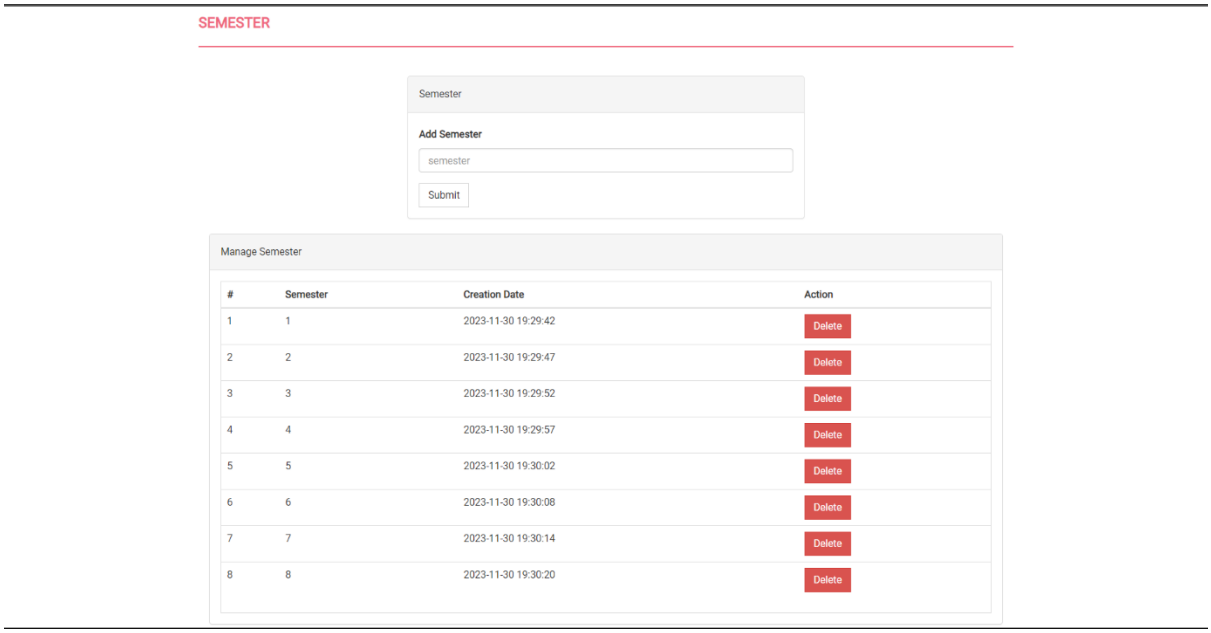
- **Functions for Administrator**



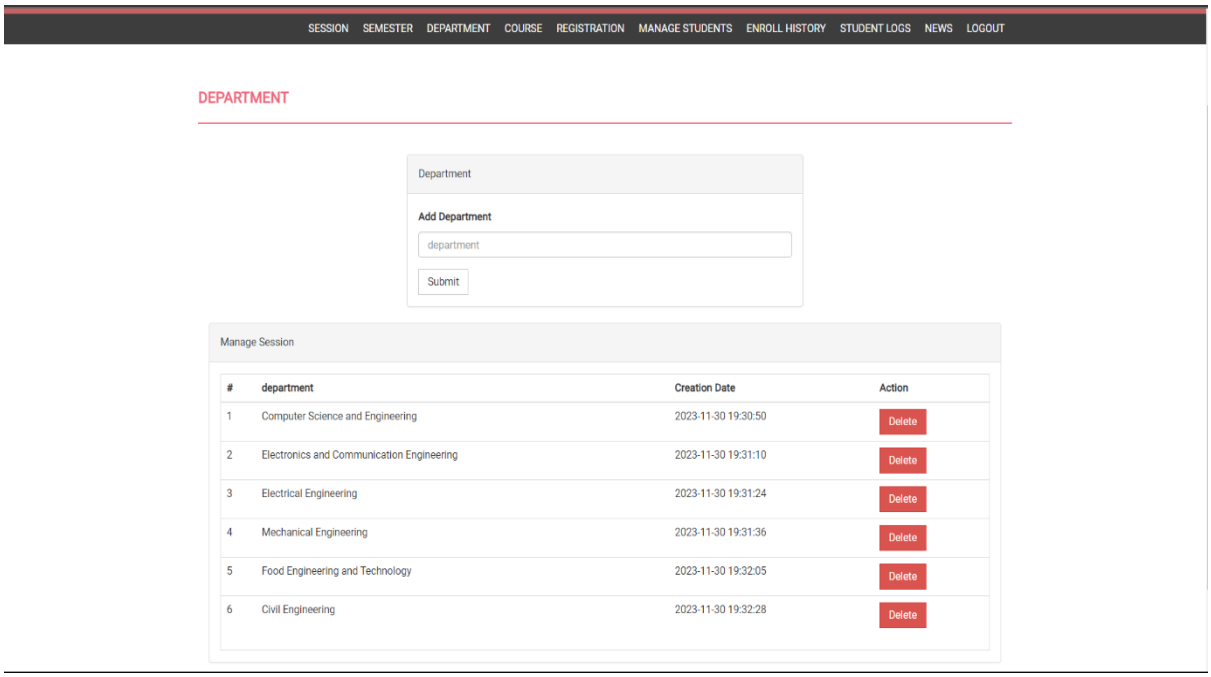
It's the login page for admin. Here the admins can login and can make changes in their own choice



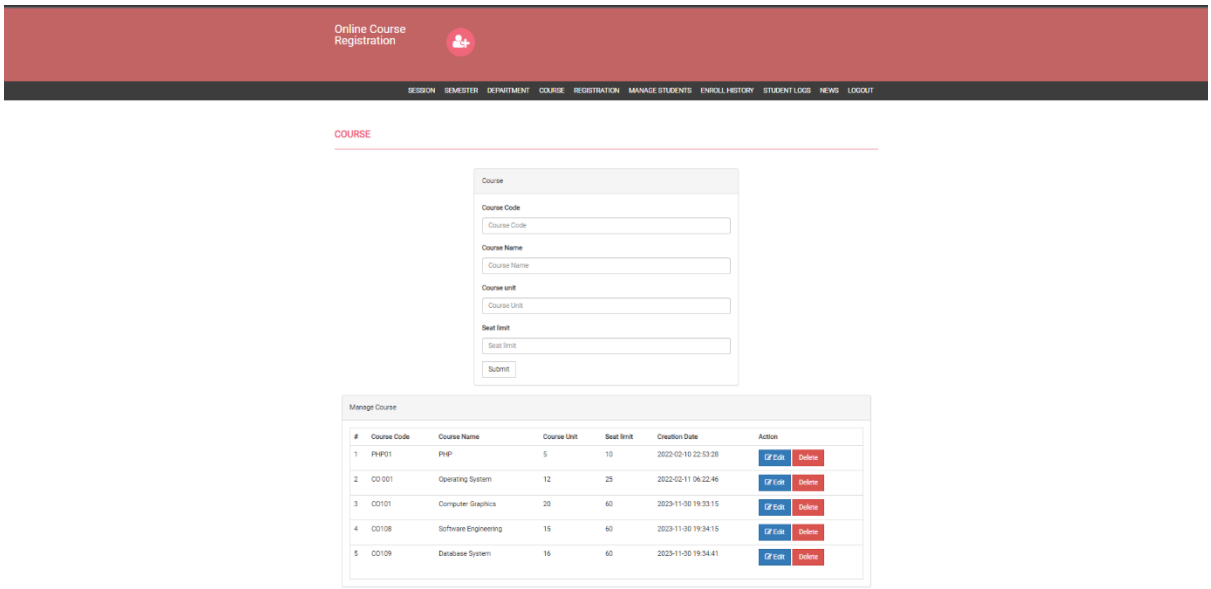
It's the session page. Here the admin can choose the session which is going on and can make/update sessions.



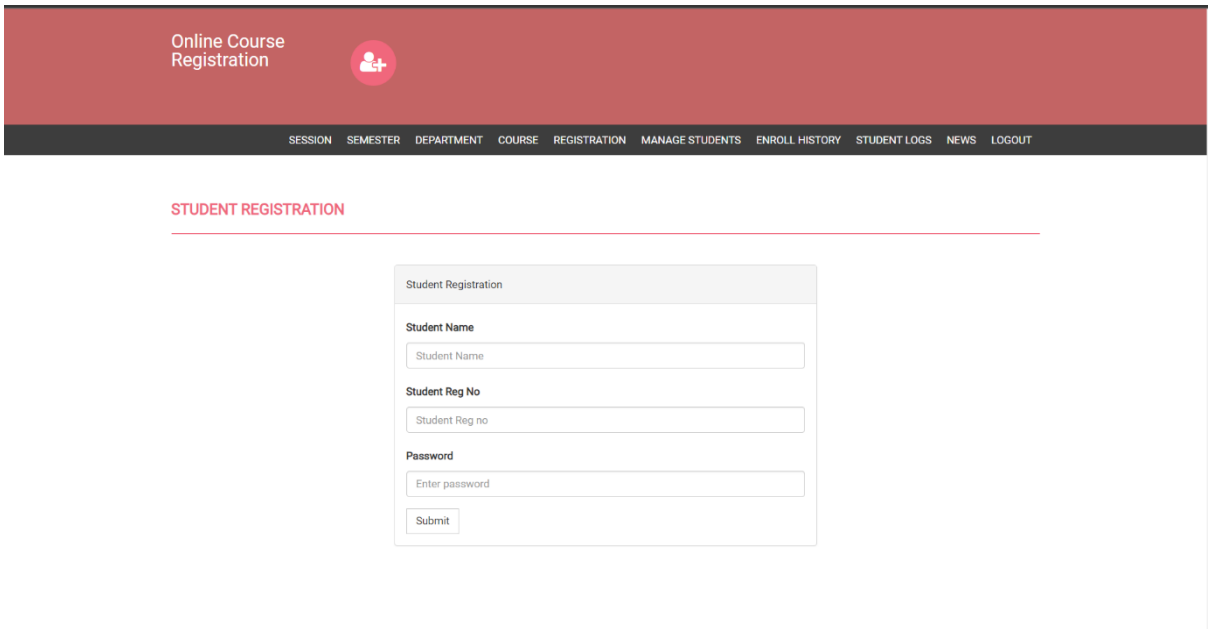
It's the page of semesters. Here the admin can choose which semester is going and can make changes accordingly.



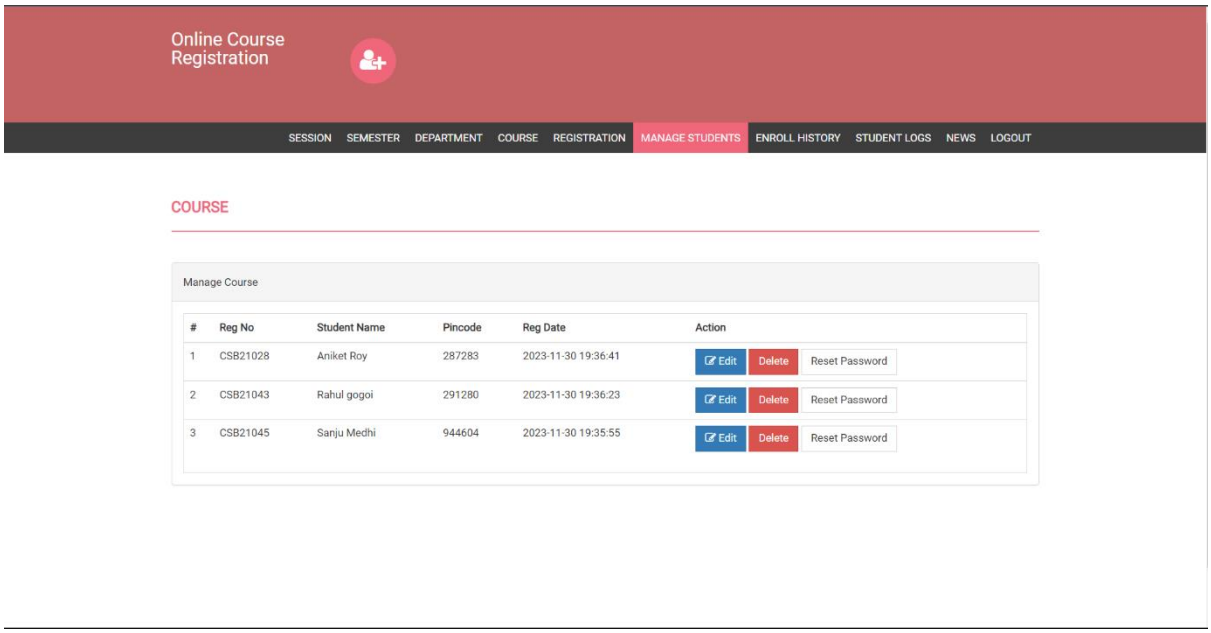
It's the page of different Departments of Engineering in Tezpur University.



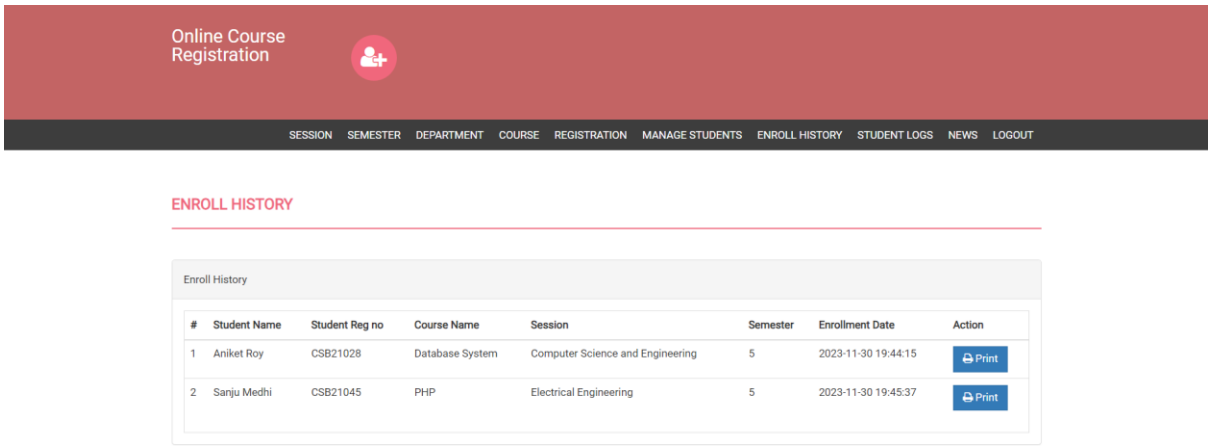
This is the page of Selection/Giving a course to the student. The admin can make changes in the course type.



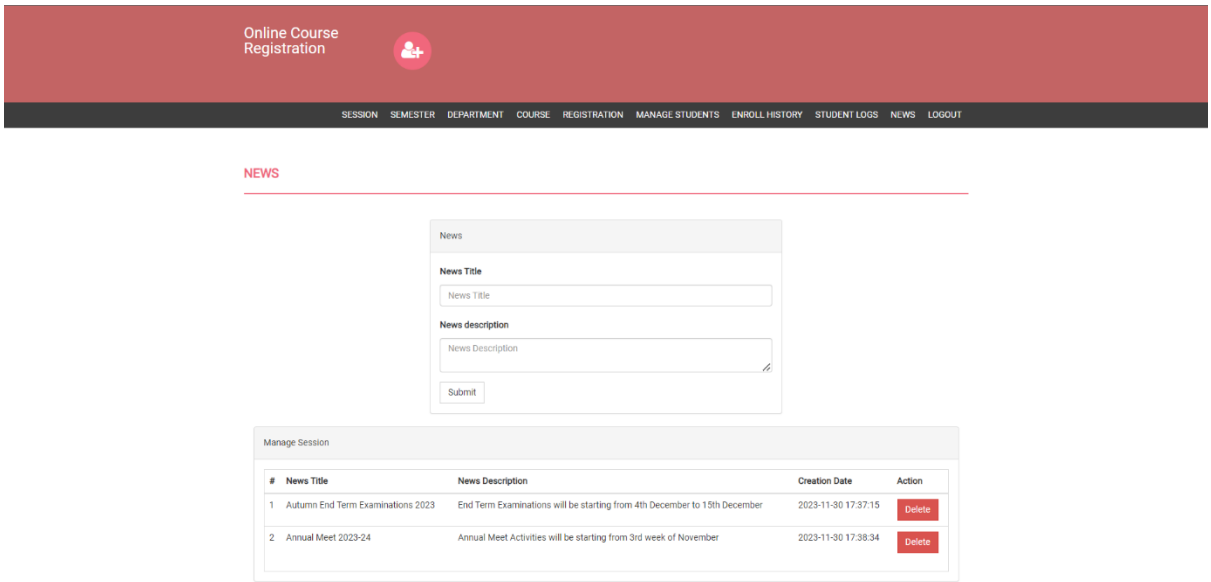
This page is for student registration. The admins can register any student



This page shows the Students who have registered the course

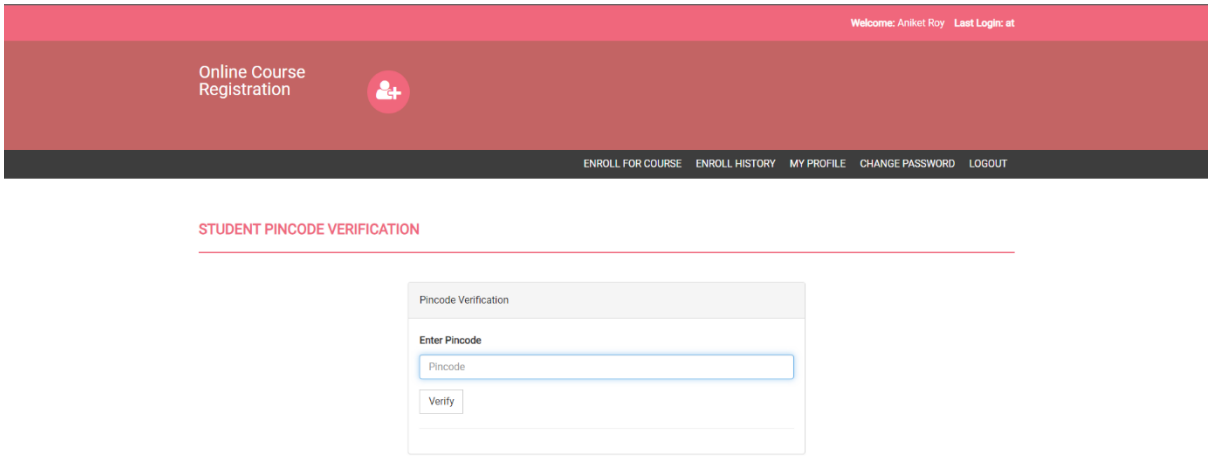


This page shows the enrolled courses to the student. The admins can make changes.

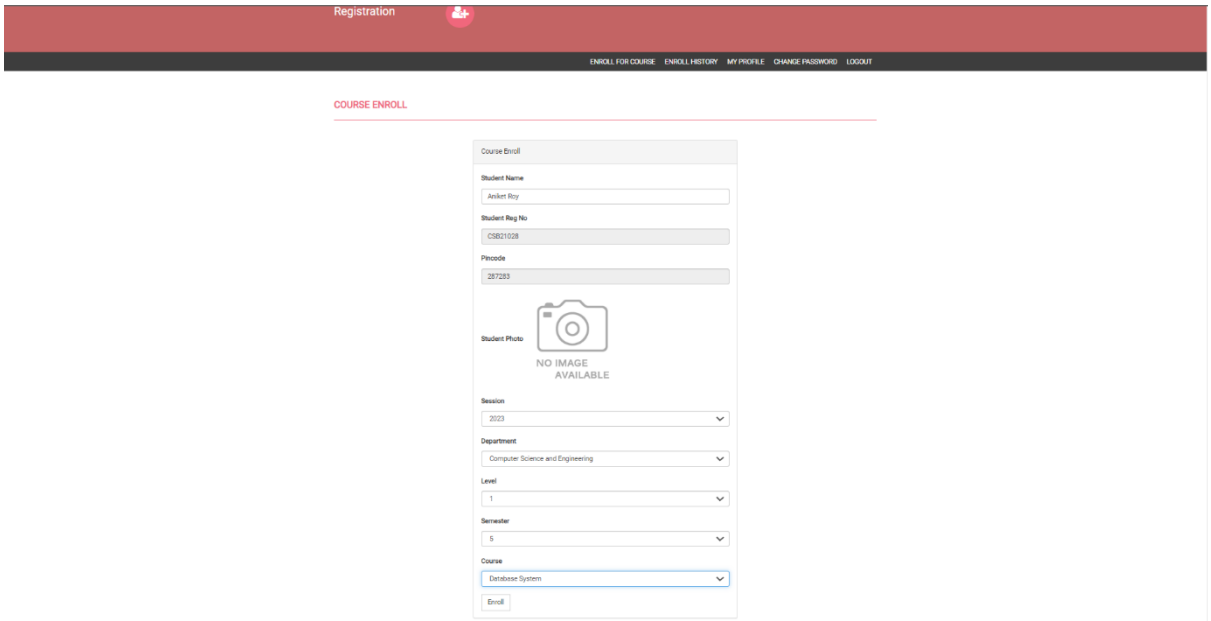


This page shows the News and Notifications. The admin can update and make changes to give all the latest news/notifications of Tezpur University

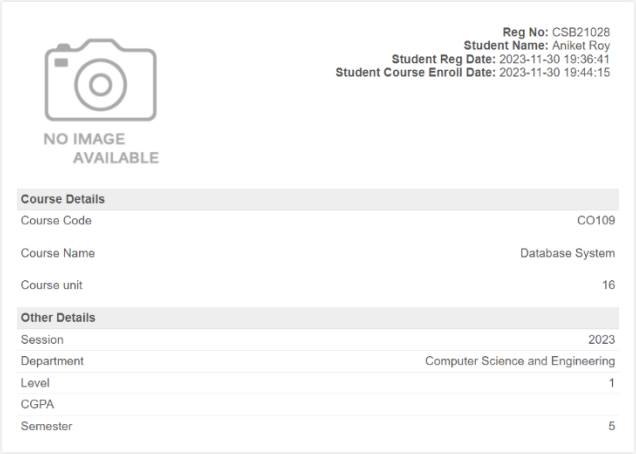
- **Functions for Students**



Here the Student have to verify the pincode which was provided by the University during registration.



This is the students course enrollment form. The student need to fill this form before the start of any semester to enroll.

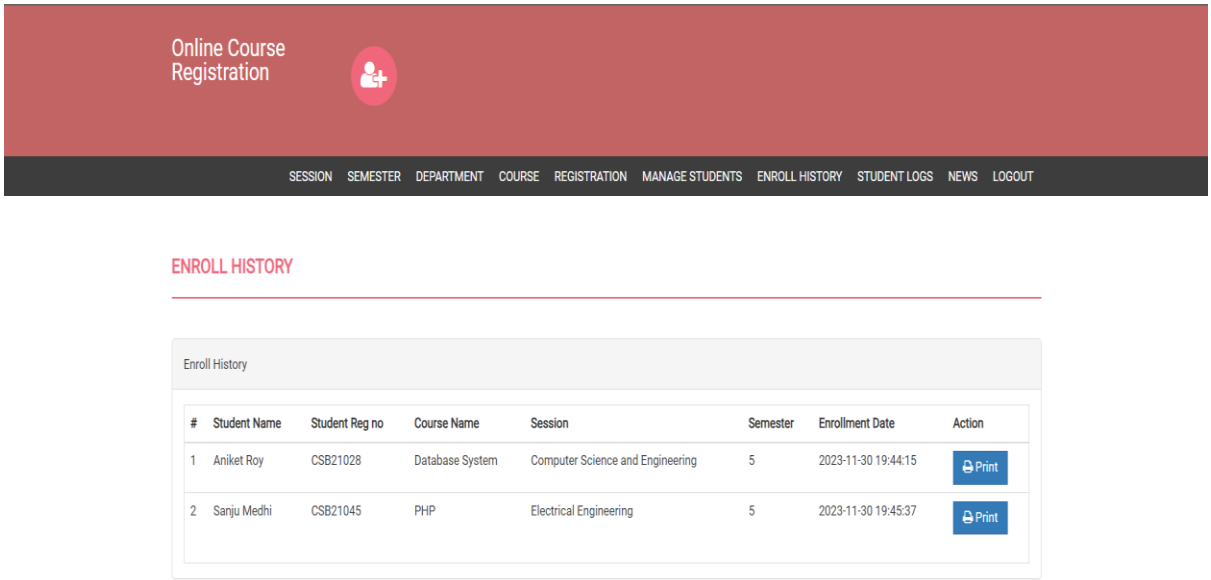


NO IMAGE AVAILABLE

Reg No: CSB21028
Student Name: Aniket Roy
Student Reg Date: 2023-11-30 19:36:41
Student Course Enroll Date: 2023-11-30 19:44:15

Course Details	
Course Code	CO109
Course Name	Database System
Course unit	16
Other Details	
Session	2023
Department	Computer Science and Engineering
Level	1
CGPA	
Semester	5

This is the printout of the course enrollment which the student makes. The student can take the printout of the following after the completion of the enrollment process.



Online Course Registration

SESSION SEMESTER DEPARTMENT COURSE REGISTRATION MANAGE STUDENTS ENROLL HISTORY STUDENT LOGS NEWS LOGOUT

ENROLL HISTORY

Enroll History

#	Student Name	Student Reg no	Course Name	Session	Semester	Enrollment Date	Action
1	Aniket Roy	CSB21028	Database System	Computer Science and Engineering	5	2023-11-30 19:44:15	Print
2	Sanju Medhi	CSB21045	PHP	Electrical Engineering	5	2023-11-30 19:45:37	Print

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This is the enrollment history of students after the completion of the process.

Conclusion and Future Development

Implementing the TU Online Student Course Registration System, the registration procedure has been simplified. Previously student had to go door to door in order to get the documents acknowledged from the concerned officials whereas the currently developed system offers an efficient way to perform these operations. The students can access the registration portal online either from a computer or a smart phone, and fill the necessary information and submit it for further approval. This web application provides us with ease of access, user friendliness and transparency. On the other hand, from organizations viewpoint, it helps in maintaining transparency, data consistency, data accessibility and easy maintenance.

Further, in future we would like to recommend certain valuable additions to the project. The most important is allowing students to submit the fees online. This can be achieved using the services of online payment system portal. This addition would further reduce the student requiring, visiting the college premises physically. It would completely automate the given system. Also, a system to digitally sign the documents can also be implemented in future. This would enable the faculty and registrar to acknowledge the documents more securely and reduce the security risk to valuable personal information. Entering student grades at semester end is also recommended to be implemented in the future. Finally, we can say that the Tezpur University Online Student Course Registration System for Student is ready for implementation and will surely provide the students and every entity associated with it a better experience. Although the project is complete and ready for implementation there is always room for improvement. As this system depends upon computer hardware, and with more and more student entries into the database there will be constant need to update server storage. To make the system more secure, In future digital acknowledgement signatures and SSL encryption can be implemented.

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